Dictionary

In [1]:
```
d1 = {}
d2 = dict()
print(d1, d2)
print(type(d1), type(d2))
```

```
{} {}
<class 'dict'> <class 'dict'>
```

In [3]:
```
d1 = {1:"One", 2:"Two", 3:"Three"}
```

In [5]:
```
print(d1)
```

```
{1: 'One', 2: 'Two', 3: 'Three'}
```

In [7]:
```
d1[1]
```

Out[7]:
```
'One'
```

In [9]:
```
d1[3]
```

Out[9]:
```
'Three'
```

In [11]:
```
d1[4]
```

```
---------------------------------------------------------------------------
KeyError                                  Traceback (most recent call last)
Cell In[11], line 1
----> 1 d1[4]

KeyError: 4
```

In [13]:
```
d1[4] = "Four"
```

In [15]:
```
d1[5] = "Five"
```

In [17]:
```
d1
```

Out[17]:
```
{1: 'One', 2: 'Two', 3: 'Three', 4: 'Four', 5: 'Five'}
```

In [19]:
```
d1[3] = "ABC"
```

In [21]:
```
d1
```

Out[21]:
```
{1: 'One', 2: 'Two', 3: 'ABC', 4: 'Four', 5: 'Five'}
```

In [23]:
```
for i in d1:
    print(i)
```

```
1
2
3
4
5
```

In [25]:
```
for i in d1:
    print(i, d1[i])
```

```
1 One
2 Two
3 ABC
4 Four
5 Five
```

Dictionary Creation Methods

In [28]:
```
li = [(1,"One"), (2,"Two"), (3,"Three"), (4,"Four")]
```

In [30]:
```
d2 = dict(li)
```

In [32]:
```
d2
```

Out[32]:
```
{1: 'One', 2: 'Two', 3: 'Three', 4: 'Four'}
```

In [34]:
```
l1 = [1,2,3,4]
l2 = ["One", "Two", "Three", "Four"]
```

In [36]:
```
zip(l1,l2)
```

Out[36]:
```
<zip at 0x1eab8f43380>
```

In [38]:
```
dict(zip(l1,l2))
```

Out[38]:
```
{1: 'One', 2: 'Two', 3: 'Three', 4: 'Four'}
```

In [40]:
```
l1 = [1,2,3,4]
l2 = ["One", "Two"]
dict(zip(l1,l2))
```

Out[40]:
```
{1: 'One', 2: 'Two'}
```

In [42]:
```
l1 = [1]
l2 = ["One", "Two", "Three", "Four"]
dict(zip(l1,l2))
```

Out[42]:
```
{1: 'One'}
```

In [44]:
```
l3 = ["One", "Two", "Three", "Four"]
```

In [46]:
```
enumerate(l3)
```

Out[46]:
```
<enumerate at 0x1eab71f5f30>
```

In [48]:
```
dict(enumerate(l3))
```

Out[48]:
```
{0: 'One', 1: 'Two', 2: 'Three', 3: 'Four'}
```

In [50]:
```
dict(enumerate(l3, start=1))
```

Out[50]:
```
{1: 'One', 2: 'Two', 3: 'Three', 4: 'Four'}
```

In [52]:
```
dict(enumerate(l3, start=101))
```

Out[52]:
```
{101: 'One', 102: 'Two', 103: 'Three', 104: 'Four'}
```

Dictionary Comprehension

In [55]:
```
li = [(1,"One"), (2,"Two"), (3,"Three"), (4,"Four")]
```

In [57]:
```
{i:j for i,j in li}
```

Out[57]:
```
{1: 'One', 2: 'Two', 3: 'Three', 4: 'Four'}
```

In [59]:
```
l1 = [1,2,3,4]
l2 = ["One", "Two", "Three", "Four"]
```

In [61]:
```
{i:j for i,j in zip(l1,l2)}
```

Out[61]:
```
{1: 'One', 2: 'Two', 3: 'Three', 4: 'Four'}
```

In [63]:
```
l3 = ["One", "Two", "Three", "Four"]
```

In [65]:
```
{i:j for i,j in enumerate(l3,start=1)}
```

Out[65]:
```
{1: 'One', 2: 'Two', 3: 'Three', 4: 'Four'}
```

Dictionary Methods

keys(), values(), items(), get(key, alt_value), setdefault(key, alt_value)

In [69]:
```
d1 = {1:"One", 2:"Two", 3:"Three", 4:"Four"}
```

In [71]:
```
d1.keys()
```

Out[71]:
```
dict_keys([1, 2, 3, 4])
```

In [73]:
```
list(d1.keys())
```

Out[73]:
```
[1, 2, 3, 4]
```

In [75]:
```
for i in d1.keys():
    print(i)
```

```
1
2
3
4
```

In [77]:
```
for i in d1.keys():
    print(i, d1[i])
```

```
1 One
2 Two
3 Three
4 Four
```

In [79]:
```
d1.values()
```

Out[79]:
```
dict_values(['One', 'Two', 'Three', 'Four'])
```

In [81]:
```
list(d1.values())
```

Out[81]:
```
['One', 'Two', 'Three', 'Four']
```

In [83]:
```
for i in d1.values():
    print(i)
```

```
One
Two
Three
Four
```

In [85]:
```
d1.items()
```

Out[85]:
```
dict_items([(1, 'One'), (2, 'Two'), (3, 'Three'), (4, 'Four')])
```

In [87]:
```
list(d1.items())
```

Out[87]:
```
[(1, 'One'), (2, 'Two'), (3, 'Three'), (4, 'Four')]
```

In [89]:
```
for i,j in d1.items():
    print(i,j)
```

```
1 One
2 Two
3 Three
4 Four
```

In [91]:
```
d1[2]
```

Out[91]:
```
'Two'
```

In [93]:
```
d1.get(2)
```

Out[93]:
```
'Two'
```

In [95]:
```
d1[5]
```

```
---------------------------------------------------------------------------
KeyError                                  Traceback (most recent call last)
Cell In[95], line 1
----> 1 d1[5]

KeyError: 5
```

In [97]:
```
d1.get(5)
```

In [99]:
```
d1.get(5,"NA")
```

Out[99]: 'NA'

In [101…
```
d1
```

Out[101…
```
{1: 'One', 2: 'Two', 3: 'Three', 4: 'Four'}
```

In [103…
```
d1.setdefault(3)
```

Out[103…
```
'Three'
```

In [105…
```
d1.setdefault(3,"Missing")
```

Out[105…
```
'Three'
```

In [107…
```
d1.setdefault(5)
```

In [109…
```
d1
```

Out[109…
```
{1: 'One', 2: 'Two', 3: 'Three', 4: 'Four', 5: None}
```

In [111…
```
d1.setdefault(6,"Unknown Value")
```

Out[111…
```
'Unknown Value'
```

In [113…
```
d1
```

Out[113…
```
{1: 'One', 2: 'Two', 3: 'Three', 4: 'Four', 5: None, 6: 'Unknown Value'}
```

In [115…
```
4 in d1
```

Out[115…
```
True
```

In [117…
```
"Two" in d1
```

Out[117…
```
False
```

In [119…
```
"Two" in d1.values()
```

Out[119…
```
True
```

In [121…
```
2 in d1.values()
```

Out[121…
```
False
```

update(dictionary/ iterable pair/keyword arguments), fromkeys(sequence, default), copy()

pop(key, alt_value), popitem(), clear()

In [124…
```
d1 = {1: 'One', 2: 'Two', 3: 'Three', 4: 'Four'}
```

In [126…
```
d2 = {5: "Five", 6:"Six"}
```

In [128…
```
d1.update(d2)
```

In [130…
```
d1
```

Out[130…
```
{1: 'One', 2: 'Two', 3: 'Three', 4: 'Four', 5: 'Five', 6: 'Six'}
```

In [132…
```
d1.update([(7,"Seven")])
```

In [134…
```
d1
```

Out[134…
```
{1: 'One', 2: 'Two', 3: 'Three', 4: 'Four', 5: 'Five', 6: 'Six', 7: 'Seven'}
```

In [136…
```
d1.update(8="Eight",9="nine")
```

```
  Cell In[136], line 1
    d1.update(8="Eight",9="nine")
              ^
SyntaxError: expression cannot contain assignment, perhaps you meant "=="?
```

In [138…
```
d1.update(_8="Eight",_9="nine")
```

In [140…
```
d1
```

Out[140…
```
{1: 'One',
 2: 'Two',
 3: 'Three',
 4: 'Four',
 5: 'Five',
 6: 'Six',
 7: 'Seven',
 '_8': 'Eight',
 '_9': 'nine'}
```

In [142…
```
l1 = ["One","Two","Three", "Four"]
```

In [144…
```
dict.fromkeys(l1)
```

Out[144…
```
{'One': None, 'Two': None, 'Three': None, 'Four': None}
```

In [146…
```
dict.fromkeys(l1,"Not defined")
```

Out[146…
```
{'One': 'Not defined',
 'Two': 'Not defined',
 'Three': 'Not defined',
 'Four': 'Not defined'}
```

In [148…
```
d2 = d1.copy()
```

In [150…
```
d1
```

Out[150…
```
{1: 'One',
 2: 'Two',
 3: 'Three',
 4: 'Four',
 5: 'Five',
 6: 'Six',
 7: 'Seven',
 '_8': 'Eight',
 '_9': 'nine'}
```

In [152…
```
d1.pop('_8')
```

Out[152…
```
'Eight'
```

In [154…
```
print(d1)
```

```
{1: 'One', 2: 'Two', 3: 'Three', 4: 'Four', 5: 'Five', 6: 'Six', 7: 'Seven', '_
9': 'nine'}
```

In [156…
```
d1.pop()
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[156], line 1
----> 1 d1.pop()

TypeError: pop expected at least 1 argument, got 0
```

In [158…
```
d1.pop('_9')
```

Out[158…
```
'nine'
```

In [160…
```
d1.pop(10)
```

```
---------------------------------------------------------------------------
KeyError                                  Traceback (most recent call last)
Cell In[160], line 1
----> 1 d1.pop(10)

KeyError: 10
```

In [162…
```
d1.pop(10,"NA")
```

Out[162…
```
'NA'
```

In [164…
```
d1
```

Out[164…
```
{1: 'One', 2: 'Two', 3: 'Three', 4: 'Four', 5: 'Five', 6: 'Six', 7: 'Seven'}
```

In [166…
```
d1.popitem()
```

Out[166…
```
(7, 'Seven')
```

In [168…
```
d1
```

Out[168…
```
{1: 'One', 2: 'Two', 3: 'Three', 4: 'Four', 5: 'Five', 6: 'Six'}
```

In [170…
```
d1.popitem()
```

Out[170… (6, 'Six')

In [172…
```
d1
```

Out[172… {1: 'One', 2: 'Two', 3: 'Three', 4: 'Four', 5: 'Five'}

In [174…
```
d1.clear()
```

In [176…
```
d1
```

Out[176… {}

In [178…
```
d1.popitem()
```

```
---------------------------------------------------------------------------
KeyError                                  Traceback (most recent call last)
Cell In[178], line 1
----> 1 d1.popitem()

KeyError: 'popitem(): dictionary is empty'
```

Frequency Count Dictionary

In [181…
```
li = [10,20,30,40,10,10,10,10,20,50,40,40,40,40]
```

In [183…
```
d = {}
for i in li:
    if i not in d:
        d[i] = 1
    else:
        d[i] += 1
print(d)
```

```
{10: 5, 20: 2, 30: 1, 40: 5, 50: 1}
```

In [185…
```
d1 = {}
for i in li:
    d1[i] = li.count(i)
print(d1)
```

```
{10: 5, 20: 2, 30: 1, 40: 5, 50: 1}
```

Invert Dictionary

In [188…
```
d1 = {1: 'One', 2: 'Two', 3: 'Three', 4: 'Four', 5: 'Five'}
```

In [190…
```
inv_di = {}
for k,v in d1.items():
    inv_di[v] = k
print(inv_di)
```

```
{'One': 1, 'Two': 2, 'Three': 3, 'Four': 4, 'Five': 5}
```